

THERESIA SAUMU

Applied AI/ML Engineer

Nairobi, Kenya (GMT+3, ~1hr ahead US East) | Open to Remote

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AI/ML Engineer building end-to-end machine learning and AI systems from **ambiguous problems to deployed solutions**. Built SolarSense, a Python and an API gateway integrating three ML microservices across 20 African cities, placing top 15 of 70 teams. Independently built StudyGapAI, an LLM-powered diagnostic platform, in one week and won 1st place at the WUD AI Hackathon 2025. Currently building production ML pipelines at FlyRank AI and AnalystLab Africa using Python, FastAPI, LangGraph, scikit-learn, and Docker.

EXPERIENCE

Machine Learning Intern | FlyRank AI (Remote) | Jun 2026 - Present

- Building **ML pipelines across large-scale search intelligence datasets** to surface content opportunities and search trends, owning the full lifecycle from problem framing through data auditing, baseline modelling, and validation.
- Applying **production-oriented engineering practices and strict data governance** to turn raw search data into evidence-based decision support for content strategy.

Data Science Intern | AnalystLab Africa (Remote) | Jul 2026 - Present

- Building **production-oriented ML systems using clean architecture and SOLID principles**, designing reproducible workflows from problem definition through deployment-ready code.
- Covering the **complete analytics lifecycle** (EDA, feature engineering, modelling, validation) with engineering discipline built for teams that need to trust and extend the code later.

KEY PROJECTS: AMBIGUITY TO DEPLOYED SYSTEM

SolarSense: Solar Viability Platform for African SMEs · Live on Railway · Top 15/70

- Architected and deployed a unified **Flask API gateway serving three independent ML microservices** for SMEs across 20 Sub-Saharan African cities, compressing viability assessment from 1–3 weeks to ~3 minutes.
- **Absorbed full ML-to-backend integration** as technical lead when the assigned backend developer stepped back, shipping centralized API key auth and Unicorn/Railway rate limiting (100 requests/minute) to keep the 19-member team's capstone on track.

StudyGapAI: AI Learning Diagnostic Platform · 1st Place, WUD AI Hackathon 2025

- **Built and deployed a full-stack AI diagnostics platform** (React, Flask, Supabase, Gemini API) solo, in one week, as the sole technical developer, from an ambiguous educational-diagnosis problem to a working product.
- **Engineered a 287-line system prompt enforcing strict JSON schema compliance** for Gemini 2.0 Flash, guaranteeing reliable structured output within a 3–5 second latency threshold.

Alzheimer's MRI Classifier: Data Integrity Investigation

- Ran an architecture search (DenseNet121 vs ResNet50 vs EfficientNetB0) and applied **Block-Wise Fine-Tuning with 140 layers frozen** on DenseNet121, reaching 99.83% accuracy before identifying a data integrity issue that had inflated the result.
- Rejected the result rather than reporting a false win, tracing it to the dataset's disclosed pre-augmentation of ~3,000 source scans into 44,000 images, and recommended a rerun of the pipeline on verified ADNI/OASIS clinical data.

TECHNICAL SKILLS

Languages: Python (primary), SQL, JavaScript

AI / LLM Engineering: LangGraph, prompt engineering, JSON schema enforcement, Google Gemini API, Mistral API

ML / Data Science: Scikit-learn, TensorFlow/Keras, XGBoost, Pandas, NumPy, feature engineering, model evaluation, EDA, data integrity auditing

Deployment / MLOps: FastAPI, Flask API gateways, Unicorn, Railway, microservices, Git/GitHub, Pytest, Docker

Also: React 18 + Vite + Tailwind, Supabase, Enterprise-level Power BI

EDUCATION

Women Techsters Data Science & Engineering Fellowship, Tech4Dev (Africa) | 2025–2026

- Selected from 14,000+ applicants (11% acceptance). Technical Lead, 19-member cross-track team; capstone placed top 15 of 70 teams.

Electrical and Electronic Engineering and Information Technology, University of Johannesburg, South Africa | 2019–2020

- Two years completed; **10 Module Distinctions**; programme not finished due to financial constraints

Certifications: IBM Data Science Professional; Google Data Analytics Professional (Coursera, 2026)